

Point-slope form



- 1 State what form is the equation $y = 3x + 6$ written in.
- 2 Write down the point-slope form of the equation of a non-vertical line with slope m that passes through the point (x_1, y_1) .
- 3 Given a straight line with slope 1 going through point $A(2, 4)$, is the slope between A and any other point on the line always 1?
- 4 For each of the following:
 - i By substituting into the equation $y = mx + b$, find the value of b for this line.
 - ii Hence, write down the equation of the line in slope-intercept form.
 - iii Find the x -intercept of the line.
 - iv Sketch the graph of the line.
 - a A line has slope $\frac{2}{3}$ and passes through the point $(-6, 2)$.
 - b A line has slope -4 and passes through the point $(3, -8)$.
- 5 A line has slope 2 and passes through the point $(-1, 5)$.
 - a By substituting into the equation $y = mx + b$, find the value of b for this line.
 - b Hence, write down the equation of the line in slope-intercept form.
 - c Use the point-slope formula to arrive at the same equation from part (b). Show your work.
 - d Sketch the graph of the line.
- 6 A line has a slope of -10 and intersects the y -axis at -1 .
 - a Use the point-slope formula to find the equation of the line in standard form.
 - b Does the point $(-7, 68)$ lie on this line?
- 7 Use the point-slope formula to find the equation of the following lines in standard form.
 - a A line passes through the point $(3, -5)$, and has a slope of $-\frac{9}{2}$.
 - b A line has a slope of $\frac{5}{9}$ and intersects the x -axis at -10 .

- 8 $A(-1, 4)$ is a point on a straight line.



If the slope of the line is 4, determine whether the following statements are correct.

a $\frac{x+1}{y-4} = 4$

b $\frac{y-4}{x+1} = 4$

c $\frac{y+1}{x-4} = 4$

d $\frac{4-y}{-1-x} = 4$

- 9 Each of the following pairs of points define a line:

i Find the slope of the line.

ii Find the equation of the line in slope-intercept form.

a $(0, 5)$ and $(5, 25)$

b $(0, -9)$ and $(3, 12)$

c $(0, 3)$ and $(-7, 31)$

d $(0, 8)$ and $(8, 80)$

e $(0, -6)$ and $(-10, -56)$

f $(6, 3)$ and $(9, 7)$

g $(1, 3)$ and $(-3, -13)$

h $(2, -7)$ and $(-5, 6)$

- 10 A line passes through the points $A(3, 4)$ and $B(-3, -9)$.

a Find the slope of the line.

b Find the equation of the line using the point-slope formula, and any one point. Express your answer in the form $Ax + By = C$ with $A > 0$.

- 11 Using the point-slope formula, find the equation of the following lines in slope-intercept form.

a A line passes through $(-2, -9)$ and has a slope of -2 .

b A line passes through $\left(\frac{7}{8}, -5\right)$ and has a slope of 2 .

c A line passes through $(7, 6)$, and has a slope of $\frac{9}{2}$.

d A line passes through $(-4, -2)$, and has a slope of 3 .

e A line passes through $(-6, 1)$ and has a slope of 4 .

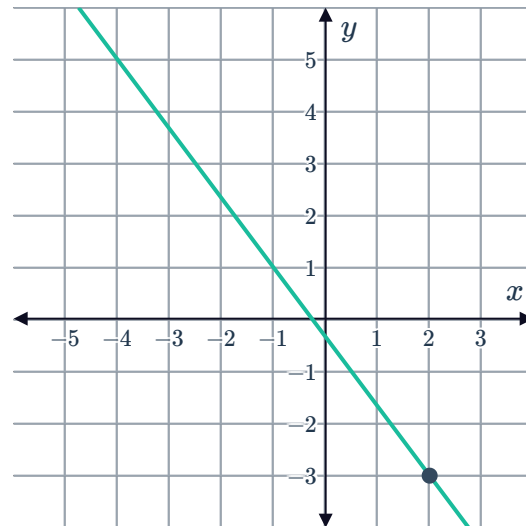
f A line passes through $\left(1, -\frac{3}{4}\right)$ and has a slope of 3 .

g A line passes through $(-5, 9)$ and has a slope of $\frac{3}{2}$.



Additional questions

- 12 State the equation of the given linear function in point-slope form.



- 13 Sketch the graph of the lines on the coordinate plane given the following equations:

a $y + 8 = 3(x + 2)$

b $y + 1 = -2(x + 1)$

c $y + 6 = -\frac{1}{3}(x - 3)$

d $y + 2 = \frac{1}{4}x$

- 14 Sally receives an order to make key chains for an upcoming festival. She is told to make at least 50 and at most 100 key chains, and will be paid \$8 for each key chain she makes. Sally makes 80 key chains and is paid \$665 altogether.
- a Let x represent the number of key chains Sally makes and y represent the amount she is paid. Write an equation in point-slope form relating x and y .
 - b Convert the equation from part (a) to slope-intercept form.
 - c Determine the flat fee that Sally was paid in addition to the \$8 per key chain that she received.
 - d State the domain of this linear relationship



- 15 Gary is flying in a plane which is moving directly away from his home at a constant speed. At 2 hours into the flight, he checked that he was 1042 miles from his home. He checks again at 7 hours into the flight to find that he is now 3542 miles from his home.
- Find the speed of the plane, in miles per hour.
 - Write an equation in point-slope form relating the number of hours flying, x , to the distance Gary is from his home, y , in miles.
 - If the destination of Gary's flight is 5542 miles from his home, find its duration.
- 16 Create a scenario where you would choose to use each of the following forms and explain why you would use that form over the others:
- Point-slope form, $(y - y_1) = m(x - x_1)$
 - Slope-intercept form, $y = mx + b$
 - Standard form, $Ax + By = C$
- 17 Rewrite the point-slope form of a linear equation in slope-intercept form. Draw connections between the two forms.
- 18 A cup is placed under a faucet which is leaking at a rate of 2 oz per hour. After 4 hours, the cup contains 15 oz of water.
- Write an equation in point-slope form relating the volume in the cup, y , to the amount of time that has passed in hours, x .
 - Now, rewrite the equation into slope-intercept form.
 - State and interpret the y -intercept of the linear equation.
 - Suppose that the cup has a maximum capacity of 18 oz. Determine how long it can be under the faucet before it starts to over flow. Now, state the domain and range.
- 19 Paige is planning an event and the brochure for the caterer at the venue shows the following graph for the meal costs.
- State and interpret the domain of the linear function.
 - Paige concludes that the unit cost per person is \$16.25. Determine if this is correct. If so, explain how to calculate this value. If not, explain the error and state the correct value.
 - State the equation of the linear function in point-slope form

